

CHAPTER 16—REGRESSION ANALYSIS: MODEL BUILDING

MULTIPLE CHOICE

1. A model in the form of $y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_p z_p + \varepsilon$ where each independent variable z_j (for $j = 1, 2, \dots, p$) is a function of x_j . x_j is known as the
 - a. general linear model
 - b. general curvilinear model
 - c. multiplicative model
 - d. multiplicative curvilinear model
2. A test used to determine whether or not first order autocorrelation is present is
 - a. z test
 - b. t test
 - c. Chi-square test
 - d. Durbin-Watson Test
3. In multiple regression analysis, the general linear model
 - a. can not be used to accommodate curvilinear relationships between dependent variables and independent variables
 - b. can be used to accommodate curvilinear relationships between the independent variables and dependent variable
 - c. must contain more than 2 independent variables
 - d. None of these alternatives is correct.
4. The following model

$$Y = \beta_0 + \beta_1 X_1 + \varepsilon$$

is referred to as a

- a. curvilinear model
 - b. curvilinear model with one predictor variable
 - c. simple second-order model with one predictor variable
 - d. simple first-order model with one predictor variable
5. In multiple regression analysis, the word linear in the term "general linear model" refers to the fact that
 - a. $\beta_0, \beta_1, \dots, \beta_p$, all have exponents of 0
 - b. $\beta_0, \beta_1, \dots, \beta_p$, all have exponents of 1
 - c. $\beta_0, \beta_1, \dots, \beta_p$, all have exponents of at least 1
 - d. $\beta_0, \beta_1, \dots, \beta_p$, all have exponents of less than 1
 6. Serial correlation is
 - a. the correlation between serial numbers of products
 - b. the same as autocorrelation
 - c. the same as leverage
 - d. None of these alternatives is correct.
 7. The joint effect of two variables acting together is called
 - a. autocorrelation
 - b. interaction
 - c. serial correlation
 - d. joint regression

8. A test to determine whether or not first-order autocorrelation is present is
 - a. a t test
 - b. an F test
 - c. a test of interaction
 - d. a chi-square test
9. Which of the following tests is used to determine whether an additional variable makes a significant contribution to a multiple regression model?
 - a. a t test
 - b. a Z test
 - c. an F test
 - d. a chi-square test
10. A variable such as Z , whose value is $Z = X_1X_2$ is added to a general linear model in order to account for potential effects of two variables X_1 and X_2 acting together. This type of effect is
 - a. impossible to occur
 - b. called interaction
 - c. called multicollinearity effect
 - d. called transformation effect
11. The following regression model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \varepsilon$$

is known as

- a. first-order model with one predictor variable
 - b. second-order model with two predictor variables
 - c. second-order model with one predictor variable
 - d. None of these alternatives is correct.
12. The parameters of nonlinear models have exponents
 - a. larger than zero
 - b. larger than 1
 - c. larger than 2
 - d. larger than 3
13. All the variables in a multiple regression analysis
 - a. must be quantitative
 - b. must be either quantitative or qualitative but not a mix of both
 - c. must be positive
 - d. None of these alternatives is correct.
14. The range of the Durbin-Watson statistic is between
 - a. -1 to 1
 - b. 0 to 1
 - c. -infinity to + infinity
 - d. 0 to 4
15. The correlation in error terms that arises when the error terms at successive points in time are related is termed
 - a. leverage
 - b. multicorrelation
 - c. autocorrelation
 - d. parallel correlation

16. What value of Durbin-Watson statistic indicates no autocorrelation is present?
 - a. 1
 - b. 2
 - c. -2
 - d. 0
17. When dealing with the problem of non-constant variance, the reciprocal transformation means using
 - a. $1/X$ as the independent variable instead of X
 - b. X^2 as the independent variable instead of X
 - c. Y^2 as the dependent variable instead of Y
 - d. $1/Y$ as the dependent variable instead of Y

NARRBEGIN: Exhibit 16-1

Exhibit 16-1

In a regression analysis involving 25 observations, the following estimated regression equation was developed.

$$\hat{Y} = 10 - 18X_1 + 3X_2 + 14X_3$$

Also, the following standard errors and the sum of squares were obtained.

$$\begin{array}{lll} S_{b1} = 3 & S_{b2} = 6 & S_{b3} = 7 \\ SST = 4,800 & & SSE = 1,296 \end{array}$$

NARREND

18. Refer to Exhibit 16-1. If you want to determine whether or not the coefficients of the independent variables are significant, the critical value of t statistic at $\alpha = 0.05$ is
 - a. 2.080
 - b. 2.060
 - c. 2.064
 - d. 1.96
19. Refer to Exhibit 16-1. The coefficient of X_1
 - a. is significant
 - b. is not significant
 - c. can not be tested, because not enough information is provided
 - d. None of these alternatives is correct.
20. Refer to Exhibit 16-1. The coefficient of X_2
 - a. is significant
 - b. is not significant
 - c. can not be tested, because not enough information is provided
 - d. None of these alternatives is correct.
21. Refer to Exhibit 16-1. The coefficient of X_3
 - a. is significant
 - b. is not significant
 - c. can not be tested, because not enough information is provided
 - d. None of these alternatives is correct.

22. Refer to Exhibit 16-1. The multiple coefficient of determination is
- 0.27
 - 0.73
 - 0.50
 - 0.33
23. Refer to Exhibit 16-1. If we are interested in testing for the significance of the relationship among the variables (i.e., significance of the model) the critical value of F at $\alpha = 0.05$ is
- 2.76
 - 2.78
 - 3.10
 - 3.07
24. Refer to Exhibit 16-1. The test statistic for testing the significance of the model is
- 0.730
 - 18.926
 - 3.703
 - 1.369
25. Refer to Exhibit 16-1. The p -value for testing the significance of the regression model is
- less than 0.01
 - between 0.01 and 0.025
 - between 0.025 and 0.05
 - between 0.05 and 0.1

NARRBEGIN: Exhibit 16-2

Exhibit 16-2

In a regression model involving 30 observations, the following estimated regression equation was obtained.

$$\hat{Y} = 170 + 34X_1 - 3X_2 + 8X_3 + 58X_4 + 3X_5$$

For this model, SSR = 1,740 and SST = 2,000.

NARREND

26. Refer to Exhibit 16-2. The value of SSE is
- 3,740
 - 170
 - 260
 - 2000
27. Refer to Exhibit 16-2. The degrees of freedom associated with SSR are
- 24
 - 6
 - 19
 - 5
28. Refer to Exhibit 16-2. The degrees of freedom associated with SSE are
- 24
 - 6
 - 19
 - 5

29. Refer to Exhibit 16-2. The degrees of freedom associated with SST are
 - a. 24
 - b. 6
 - c. 19
 - d. None of these alternatives is correct.
30. Refer to Exhibit 16-2. The value of MSR is
 - a. 10.40
 - b. 348
 - c. 10.83
 - d. 52
31. Refer to Exhibit 16-2. The value of MSE is
 - a. 348
 - b. 10.40
 - c. 10.83
 - d. 32.13
32. Refer to Exhibit 16-2. The test statistic F for testing the significance of the above model is
 - a. 32.12
 - b. 6.69
 - c. 4.8
 - d. 58
33. Refer to Exhibit 16-2. The p -value for testing the significance of the regression model is
 - a. less than 0.01
 - b. between 0.01 and 0.025
 - c. between 0.025 and 0.05
 - d. between 0.05 and 0.1
34. Refer to Exhibit 16-2. The coefficient of determination for this model is
 - a. 0.6923
 - b. 0.1494
 - c. 0.1300
 - d. 0.8700

NARRBEGIN: Exhibit 16-3

Exhibit 16-3

Below you are given a partial computer output based on a sample of 25 observations.

	Coefficient	Standard Error
Constant	145	29
X_1	20	5
X_2	-18	6
X_3	4	4

NARREND

35. Refer to Exhibit 16-3. The estimated regression equation is
- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$
 - $E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$
 - $\hat{Y} = 29 + 5X_1 + 6X_2 + 4X_3$
 - $\hat{Y} = 145 + 20X_1 - 18X_2 + 4X_3$
36. Refer to Exhibit 16-3. We want to test whether the parameter β_2 is significant. The test statistic equals
- 4
 - 5
 - 3
 - 3
37. Refer to Exhibit 16-3. The critical t value obtained from the table to test an individual parameter at the 5% level is
- 2.06
 - 2.069
 - 2.074
 - 2.080

NARRBEGIN: Exhibit 16-4

Exhibit 16-4

In a laboratory experiment, data were gathered on the life span (Y in months) of 33 rats, units of daily protein intake (X_1), and whether or not agent X_2 (a proposed life extending agent) was added to the rats diet ($X_2 = 0$ if agent X_2 was not added, and $X_2 = 1$ if agent was added.) From the results of the experiment, the following regression model was developed.

$$\hat{Y} = 36 + 0.8X_1 - 1.7X_2$$

Also provided are $SSR = 60$ and $SST = 180$.

NARREND

38. Refer to Exhibit 16-4. From the above function, it can be said that the life expectancy of rats that were given agent X_2 is
- 1.7 months more than those who did not take agent X_2
 - 1.7 months less than those who did not take agent X_2
 - 0.8 months less than those who did not take agent X_2
 - 0.8 months more than those who did not take agent X_2
39. Refer to Exhibit 16-4. The life expectancy of a rat that was given 3 units of protein daily, and who took agent X_2 is
- 36.7
 - 36
 - 49
 - 38.4

40. Refer to Exhibit 16-4. The life expectancy of a rat that was not given any protein and that did not take agent X_2 is
 - a. 36.7
 - b. 34.3
 - c. 36
 - d. 38.4
41. Refer to Exhibit 16-4. The life expectancy of a rat that was given 2 units of agent X_2 daily, but was not given any protein is
 - a. 32.6
 - b. 36
 - c. 38
 - d. 34.3
42. Refer to Exhibit 16-4. The degrees of freedom associated with SSR are
 - a. 2
 - b. 33
 - c. 32
 - d. 30
43. Refer to Exhibit 16-4. The degrees of freedom associated with SSE are
 - a. 3
 - b. 33
 - c. 32
 - d. 30
44. Refer to Exhibit 16-4. The multiple coefficient of determination is
 - a. 0.2
 - b. 0.5
 - c. 0.333
 - d. 5
45. Refer to Exhibit 16-4. If we want to test for the significance of the model, the critical value of F at 95% confidence is
 - a. 4.17
 - b. 3.32
 - c. 2.92
 - d. 1.96
46. Refer to Exhibit 16-4. The test statistic for testing the significance of the model is
 - a. 0.50
 - b. 5.00
 - c. 0.25
 - d. 0.33
47. Refer to Exhibit 16-4. The p -value for testing the significance of the regression model is
 - a. less than 0.01
 - b. between 0.01 and 0.025
 - c. between 0.025 and 0.05
 - d. between 0.05 and 0.10
48. Refer to Exhibit 16-4. The model
 - a. is significant
 - b. is not significant
 - c. Not enough information is provided to answer this question.
 - d. None of these alternatives is correct.

PROBLEM

- In a regression analysis involving 21 observations and 4 independent variables, the following information was obtained.

$$r^2 = 0.80$$

$$S = 5.0$$

Based on the above information, fill in all the blanks in the following ANOVA.

Hint: $r^2 = \frac{SSR}{SST}$, but also $r^2 = 1 - \frac{SSE}{SST}$.

Source	DF	SS	MS	F
Regression	_____?	_____?	_____?	_____?
Error (Residual)	_____?	_____?	_____?	
Total	_____?	_____?		

- We are interested in determining what type of model best describes the relationship between two variables x and y .
 - For a given data set, an estimated regression equation relating x and y of the form $\hat{y} = b_0 + b_1x$ was developed, using Excel. The results are shown below. Comment on the adequacy of this equation for predicting y . Let $\alpha = .05$.

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.5095
R Square	0.2596
Adjusted R Square	0.1362
Standard Error	2.0745
Observations	8

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	9.0536	9.0536	2.1037	0.1971
Residual	6	25.8214	4.3036		
Total	7	34.875			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	2.7857	1.6164	1.7234	0.1356
x	0.4643	0.3201	1.4504	0.1971

- An estimated regression equation for the same data set (as in part a) of the form $\hat{y} = b_0 + b_1x + b_2x^2$ was developed. The Excel output is shown below. Comment on the adequacy of this equation for predicting y . Let $\alpha = .05$.

SUMMARY
OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.9680
R Square	0.9370
Adjusted R Square	0.9118
Standard Error	0.6628
Observations	8

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	32.6786	16.3392	37.1951	0.0010
Residual	5	2.1964	0.4393		
Total	7	34.875			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	-2.8393	0.9247	-3.0706	0.0278
x	3.8393	0.4714	8.1437	0.0005
x-squared	-0.375	0.0511	-7.3335	0.0007

c. Use the results of Part b and predict y when x = 4.

3. Consider the following data for two variables x and y.

x	y
1	1
4	6
7	9
8	7
9	4
10	3

a. An estimated regression equation of the form $\hat{y} = b_0 + b_1x$ was developed for the above data and the results are shown below. Comment on the adequacy of this equation for predicting y. Let $\alpha = .05$.

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.3052
R Square	0.0932
Adjusted R Square	-0.1335
Standard Error	3.0857
Observations	6

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	3.9130	3.9130	0.4110	0.5564
Residual	4	38.0870	9.5217		
Total	5	42			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	3.3043	2.9297	1.1279	0.3224
x	0.2609	0.4069	0.6411	0.5564

- b. A regression equation of the form $\hat{y} = b_0 + b_1x + b_2x^2$ was developed for the above data and results are shown below. Comment on the adequacy of this equation for predicting y. Let $\alpha = .05$.

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.9508
R Square	0.9041
Adjusted R Square	0.8401
Standard Error	1.1588
Observations	6

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	37.9713	18.9856	14.1376	0.0297
Residual	3	4.0287	1.343		
Total	5	42			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	-2.6808	1.6196	-1.655	0.1964
x	3.6803	0.6960	5.2879	0.0132
x-squared	-0.3133	0.0622	-5.036	0.0151

- c. Predict the value of y when $x = 5$.
4. The following estimated regression equation has been developed for the relationship between y, the dependent variable, and x, the independent variable.
- $$\hat{y} = 60 + 200x - 6x^2$$
- The sample size for this regression model was 23, and $SSR = 600$ and $SSE = 400$.
- Compute the coefficient of determination.
 - Using $\alpha = .05$, test for a significant relationship.
5. A data set consisting of 7 observations of a dependent variable y and two independent variables x_1 and x_2 was used in a regression analysis. Using (x_1) as the only independent variable, the following function is provided.

$$\hat{y} = 0.408 + 1.338x_1$$

The SSE for the above model is 39.535.

Using both x_1 and x_2 as independent variables yields the following function.

$$\hat{y} = 0.805 + 0.498x_1 - 0.477x_2$$

The SSE for this latter function is 1.015.

Use an F test and determine if x_2 contributes significantly to the model. Let $\alpha = 0.05$.

6. Monthly total production costs and the number of units produced at a local company over a period of 10 months are shown below.

Month	Production Costs (Y_i) (in millions \$)	Units Produced (X_i) (in millions)
1	1	2
2	1	3
3	1	4
4	2	5
5	2	6
6	4	7
7	5	8
8	7	9
9	9	10
10	12	10

- Draw a scatter diagram for the above data.
- Assume that a model in the form of

$$Y = \beta_0 + \beta_1 X_2 + \varepsilon$$

best describes the relationship between X and Y. Estimate the parameters of this curvilinear regression equation.

7. Consider the following data.

Y_i	X_i
2	1
3	4
5	6
8	7
10	8

- Draw a scatter diagram. Does the relationship between X and Y appear to be linear?
- Assume the relationship between X and Y can best be given by

$$Y = \beta_0 + \beta_1 X^2 + \varepsilon$$

Estimate the parameters of this curvilinear function.

8. Part of an Excel output relating Y (dependent variable) and 4 independent variables, X1 through X4, is shown below.

Summary Output

<i>Regression Statistics</i>	
Multiple R	?
R Square	?
Adjusted R Square	?
Standard Error	72.6093
Observations	20

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	?	422975.2376	?	?	0.0000
Residual	?	?	?		
Total	?	?			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	-203.6125	100.2940	?	0.0605
X1	0.6483	0.1110	?	0.0000
X2	0.0190	0.0065	?	0.0101
X3	40.4577	7.5940	?	0.0001
X4	-0.1032	20.7823	?	0.9961

- Fill in all the blanks marked with "?"
 - At 95% confidence, which independent variables are significant and which ones are not? Fully explain how you arrived at your answers.
9. In a regression analysis involving 20 observations and five independent variables, the following information was obtained.

ANALYSIS OF VARIANCE				
Source of Variation	Degrees of Freedom	Sum of Squares	Mean Squares	F
Regression	_____?	_____?	_____?	_____?
Error (Residual)	_____?	_____?	30	
Total		990		

Fill in all the blanks in the above ANOVA table.

10. A researcher is trying to decide whether or not to add another variable to his model. He has estimated the following model from a sample of 28 observations.

$$\hat{Y} = 23.62 + 18.86X_1 + 24.72X_2$$

$$SSE = 1,425 \quad SSR = 1,326$$

He has also estimated the model with an additional variable X_3 . The results are

$$\hat{Y} = 25.32 + 15.29X_1 + 7.63X_2 + 12.72X_3$$

$$SSE = 1,300 \quad SSR = 1,451$$

What advice would you give this researcher? Use a .05 level of significance.

11. We want to test whether or not the addition of 3 variables to a model will be statistically significant. You are given the following information based on a sample of 25 observations.

$$\hat{Y} = 62.42 - 1.836X_1 + 25.62X_2$$

$$SSE = 725 \quad SSR = 526$$

The equation was also estimated including the 3 variables. The results are

$$\hat{Y} = 59.23 - 1.762X_1 + 25.638X_2 + 16.237X_3 + 15.297X_4 - 18.723X_5$$

$$SSE = 520 \quad SSR = 731$$

- a. State the null and alternative hypotheses.
 - b. Test the null hypothesis at the 5% level of significance.
12. Multiple regression analysis was used to study the relationship between a dependent variable, Y , and three independent variables X_1 , X_2 and, X_3 . The following is a partial result of the regression analysis involving 20 observations.

	Coefficient	Standard Error
Intercept	20.00	5.00
X_1	15.00	3.00
X_2	8.00	5.00
X_3	-18.00	10.00

Analysis of Variance

Source	DF	SS	MS	F
Regression			80	
Error		320		

- a. Compute the coefficient of determination.
 - b. Perform a t test and determine whether or not β_1 is significantly different from zero ($\alpha = 0.05$).
 - c. Perform a t test and determine whether or not β_2 is significantly different from zero ($\alpha = 0.05$).
 - d. Perform a t test and determine whether or not β_3 is significantly different from zero ($\alpha = 0.05$).
 - e. At $\alpha = 0.05$, perform an F test and determine whether or not the regression model is significant.
13. Multiple regression analysis was used to study the relationship between a dependent variable, Y , and four independent variables; X_1 , X_2 , X_3 and, X_4 . The following is a partial result of the regression analysis involving 31 observations.

	Coefficient	Standard Error
Intercept	18.00	6.00
X_1	12.00	8.00
X_2	24.00	48.00
X_3	-36.00	36.00
X_4	16.00	2.00

Analysis of Variance

Source	df	SS	MS	F
Regression			125	
Error				
Total		760		

- a. Compute the coefficient of determination.
 - b. Perform a t test and determine whether or not β_1 is significantly different from zero ($\alpha = 0.05$).
 - c. Perform a t test and determine whether or not β_4 is significantly different from zero ($\alpha = 0.05$).
 - d. At $\alpha = 0.05$, perform an F test and determine whether or not the regression model is significant.
14. A regression model relating a dependent variable, Y , with one independent variable, X_1 , resulted in an SSE of 400. Another regression model with the same dependent variable, Y , and two independent variables, X_1 and X_2 , resulted in an SSE of 320. At $\alpha = .05$, determine if X_2 contributed significantly to the model. The sample size for both models was 20.

15. A regression model with one independent variable, X_1 , resulted in an SSE of 50. When a second independent variable, X_2 , was added to the model, the SSE was reduced to 40. At $\alpha = 0.05$, determine if X_2 contributes significantly to the model. The sample size for both models was 30.
16. When a regression model was developed relating sales (Y) of a company to its product's price (X_1), the SSE was determined to be 495. A second regression model relating sales (Y) to product's price (X_1) and competitor's product price (X_2) resulted in an SSE of 396. At $\alpha = 0.05$, determine if the competitor's product's price contributed significantly to the model. The sample size for both models was 33.
17. A regression model relating units sold (Y), price (X_1), and whether or not promotion was used ($X_2 = 1$ if promotion was used and 0 if it was not) resulted in the following model.

$$\hat{Y} = 120 - 0.03X_1 + 0.7X_2$$

and the following information is provided.

$$n = 15 \quad S_{b1} = .01 \quad S_{b2} = 0.1$$

- a. Is price a significant variable?
 - b. Is promotion significant?
18. A regression model relating the yearly income (Y), age (X_1), and the gender of the faculty member of a university ($X_2 = 1$ if female and 0 if male) resulted in the following information.

$$\hat{Y} = 5,000 + 1.2X_1 + 0.9X_2$$

$$n = 20 \quad SSE = 500 \quad SSR = 1,500$$

$$S_{b1} = 0.2 \quad S_{b2} = 0.1$$

- a. Is gender a significant variable?
 - b. Determine the multiple coefficient of determination.
19. A regression analysis was applied in order to determine the relationship between a dependent variable and 8 independent variables. The following information was obtained from the regression analysis.

$$R \text{ Square} = 0.80$$

$$SSR = 4,280$$

$$\text{Total number of observations } n = 56$$

- a. Fill in the blanks in the following ANOVA table.
- b. Is the model significant? Let $\alpha = 0.05$.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Squares	F
Regression	_____?	_____?	_____?	_____?
Error	_____?	_____?	_____?	
Total	_____?	_____?		

20. In a regression analysis involving 18 observations and four independent variables, the following information was obtained.

Multiple R = 0.6000
R Square = 0.3600
Standard Error = 4.8000

Based on the above information, fill in all the blanks in the following ANOVA table.

ANALYSIS OF VARIANCE				
Source of Variation	Degrees of Freedom	Sum of Squares	Mean Squares	F
Regression	_____?	_____?	_____?	_____?
Error	_____?	_____?	_____?	

21. The following are partial results of a regression analysis involving sales (Y in millions of dollars), advertising expenditures (X_1 in thousands of dollars), and number of salespeople (X_2) for a corporation. The regression was performed on a sample of 10 observations.

	Coefficient	Standard Error
Constant	50.00	20.00
X_1	3.60	1.20
X_2	0.20	0.20

- a. At $\alpha = 0.05$, test for the significance of the coefficient of advertising.
b. If the company uses \$20,000 in advertisement and has 300 salespersons, what are the expected sales? (Give your answer in dollars.)
22. A regression analysis was applied in order to determine the relationship between a dependent variable and 4 independent variables. The following information was obtained from the regression analysis.

R Square = 0.80

SSR = 680

Total number of observations $n = 45$

- a. Fill in the blanks in the following ANOVA table.
b. At $\alpha = 0.05$ level of significance, test to determine if the model is significant.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Squares	F
Regression	_____?	_____?	_____?	_____?
Error (Residual)	_____?	_____?	_____?	
Total	_____?	_____?		

23. A regression analysis (involving 45 observations) relating a dependent variable (Y) and two independent variables resulted in the following information.

$$\hat{Y} = 0.408 + 1.3387X_1 + 2X_2$$

The SSE for the above model is 49.

When two other independent variables were added to the model, the following information was provided.

$$\hat{Y} = 1.2 + 3.0X_1 + 12X_2 + 4.0X_3 + 8X_4$$

This latter model's SSE is 40.

At 95% confidence test to determine if the two added independent variables contribute significantly to the model.

24. A computer manufacturer has developed a regression model relating Sales (Y in \$10,000) with four independent variables. The four independent variables are Price (in dollars), Competitor's Price (in dollars), Advertising (in \$1000) and Type of computer produced (Type = 0 if desktop, Type = 1 if laptop). Part of the regression results are shown below.

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>
Regression	4	27641631.121	6910407.780
Residual	35	42277876.624	1207939.332

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>
Intercept	2268.233	1237.880	
Price	-0.803	0.316	
Competitor's Price	0.859	0.281	
Advertising	0.216	0.079	
Type	567.806	373.400	

- What has been the sample size?
 - Determine the coefficient of determination.
 - Compute the test statistic t for each of the four independent variables.
 - Determine the p -values for the four variables.
 - At 95% confidence, which variables are significant? Explain how you arrived at your conclusion.
 - At 95% confidence, test to see if the regression model is significant.
25. Thirty-four observations of a dependent variable (Y) and two independent variables resulted in an SSE of 300. When a third independent variable was added to the model, the SSE was reduced to 250. At 95% confidence, determine whether or not the third independent variable contributes significantly to the model.
26. Forty-eight observations of a dependent variable (Y) and five independent variables resulted in an SSE of 438. When two additional independent variables were added to the model, the SSE was reduced to 375. At 95% confidence, determine whether or not the two additional independent variables contribute significantly to the model.
27. A regression analysis was applied in order to determine the relationship between a dependent variable and 4 independent variables. The following information was obtained from the regression analysis.
R Square = 0.60
SSR = 4,800
Total number of observations $n = 35$
- Fill in the blanks in the following ANOVA table.
 - At $\alpha = 0.05$ level of significance, test to determine if the model is significant.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Squares	F
Regression	_____?	_____?	_____?	_____?
Error (Residual)	_____?	_____?	_____?	
Total	_____?	_____?		

28. A regression analysis relating a company's sales, their advertising expenditure, price, and time resulted in the following.

Regression Statistics

Multiple R	0.8800
R Square	0.7744
Adjusted R Square	0.7560
Standard Error	232.29
Observations	25

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	53184931.86	17728310.62	328.56	0.0000
Residual	21	1133108.30	53957.54		
Total	24	54318040.16			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	927.23	1229.86	0.75	0.4593
Advertising (X ₁)	1.02	3.09	0.33	0.7450
Price (X ₂)	15.61	5.62	2.78	0.0112
Time (X ₃)	170.53	28.18	6.05	0.0000

- At 95% confidence, determine whether or not the regression model is significant. Fully explain how you arrived at your conclusion (give numerical reasoning) and what your answer indicates.
 - At 95% confidence determine which variables are significant and which are not. Explain how you arrived at your conclusion (Give numerical reasoning).
 - Fully explain the meaning of R-square, which is given in this model. Be very specific and give numerical explanation.
29. Ziba, Inc. has provided the following information regarding their sales for January through December of 2009. (Part of the data file is shown below.)

	<i>Sales</i> (Y in \$100,000)	<i>Advertising</i> (X ₁ in \$10,000)	<i>Time</i> (X ₂)
Year 2009			
January			1
February			2
.....			
.....			
November	36	26	11
December	37	28	12

The results of the regression analysis relating these variables are shown below.

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	13.01	0.6334	20.5379	0.0000
Advertising (X ₁ in 10,000)	0.31	0.1293	2.3784	0.0413
Time (X ₂)	1.39	0.2863	4.8390	0.0009

- The company is planning to increase their advertising by 5% per month for the months of January and February of 2010. What would be the advertising for January and February of 2010? Give your answers in dollars.

- b. Use the regression model that is provided above and forecast sales for January and February of 2010, assuming the company increases their advertising by 5% per month for the months of January and February of 2010. Show your computations and write your answers in dollars below.